Geonwoo **Kim**

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"There is no failure, only growth."

Education

Inha University, Department of Artificial Intelligence — *Incheon, S. Korea*

B.S. in Artificial Intelligence — *Mar. 2021 – Aug. 2027 (Expected)*

- GPA: **4.16 / 4.5**.

Research Interests

Generative Models, Multimodal Learning, World Models

Research Experience

DAVIAN Lab | Kim Jaechul Graduate School of AI, KAIST — *Seoul, S. Korea*

Undergraduate Research Intern — *Jan. 2026 – Present*

- Researched **multi-view conditioned video generation**, integrating 3D-grounded data synthesis and geometric conditioning.
- Co-led the research as a **co-first author** and submitted the paper to **ICLR 2027**; advised by Professor Jaegul Choo.

Multimodal AI Lab | Inha University — *Incheon, S. Korea*

Undergraduate Researcher — *Dec. 2024 – Dec. 2025*

- Reproduced and compared **training-free subject-driven image editing** methods using reference images, target images, and masks.
- Extended masked editing to preserve subject identity and background context; advised by Professor Pilhyeon Lee.

Publications

A Paper on Multi-view Conditioned Video Generation — *ICLR 2027, under review*

Co-first author — *2026*

- Developed 3D-grounded data synthesis and geometric conditioning for **subject-consistent video generation** across viewpoints.

Project Experience

Action-Conditioned Robot World Model — *Code / Slides*

Team Leader — *Jul. 2026 – Aug. 2026*

- Led an **action-conditioned video generation** pipeline predicting future robot observations from an initial image and an action sequence.
- Adapted **Cosmos-Predict2.5-2B** with **LoRA**, an inverse-dynamics auxiliary loss, and model soup, integrating training through final inference.
- Reduced the public leaderboard score by **66.1% over the official baseline**, winning the **Top Excellence Award (2nd, graduate track)**.

LG Aimers 9th: Pitch Control Prediction — *Code*

Team Leader — *2026*

- Engineered historical pitching features to predict **KBO pitch-control success probabilities**.
- Applied residual modeling and segment-wise calibration to a **publicly available CatBoost ensemble backbone**.
- Achieved **2.09 times the official baseline's score** on the public leaderboard, ranking **88th**.

Agent Action Prediction — *Code*

Team Leader — *Jul. 2026*

- Built a **14-class next-action predictor** from agent interaction history with session-level validation to prevent leakage.
- Optimized encoder ensembles through **vocabulary pruning** and conditional inference under model-size and runtime constraints.
- Achieved a final leaderboard score of **0.78837** after controlled experiments on input serialization, augmentation, and ensemble selection.

COSFLOW: AI Automation Solution for Cosmetic OEM Companies

Website

Co-founding Member / AI Researcher — *Jun. 2025 – Sep. 2025*

- Developed an **AI Agent solution** streamlining cosmetic ingredient compliance workflows for OEM clients.
- Built an **LLM-integrated pipeline** that standardizes ingredient names, queries multi-country databases, and generates structured compliance reports (PDF).
- Implemented **vector database-based ingredient similarity search** and **GPT-4.1-driven report generation**, including ingredient cards and dashboards.

Text-Guided Image Colorization (Inha AI Challenge)

Code

Team Leader — *Jun. 2025 – Aug. 2025*

- Led an AI competition project focused on **text-guided colorization** of grayscale images.
- Built a candidate-generation pipeline combining **DDColor and FLUX** for text-conditioned colorization.
- Designed **CLIP-based candidate selection** to choose the generated image most aligned with the text description.
- Achieved the **BK21 Director's Award (Excellence)** with a final score of **0.6178**.

AI-Generated vs. Human-Written Text Classification Challenge

Seoul, S. Korea

Team Leader — *Jun. 2025 – Jul. 2025*

- Developed an AI model distinguishing **human-written and LLM-generated Korean texts**, despite full-text-level labeling.
- Improved paragraph-level discrimination via **LLM-based data augmentation** and **ensemble modeling**.
- Used **RoBERTa-Large, BERT-base, and KoBigBird** embeddings with **3:1 undersampling** to address imbalance.
- Achieved **ROC-AUC 0.906** in the classification challenge.

Enterprise-tailored AI Research & Report Generation Agent

Incheon, S. Korea

Co-founding Member / AI Researcher — *Mar. 2025 – Jun. 2025*

- Developed an **AI Agent system integrating RAG retrieval**, vector database search, and template-based report automation.
- Built a **LangChain-powered CSV Analysis Agent** supporting data queries, visualization, and automated summaries in Streamlit.
- Designed an **LLM pipeline** for domain-adaptive generation and reliability evaluation of external sources.
- Secured government funding through the **Korean Pre-Startup Package** and **Lab-based Startup Leading University Program**.

Awards & Honors

2026.10	**1st**, National Park Satellite Monitoring AI Challenge	*AIFactory*
2026.09	**Top Excellence Award (2nd, graduate track)**, Robot World Model AI Challenge	*Inha Univ.*
2026.05	**Excellence Award (3rd)**, Inter Campus AI Challenge 2026	*Upstage*
2025.09	**Director's Award (Excellence, 3rd)**, Incheon Startup Competition	*Incheon*
2025.08	**BK21 Director's Award (Excellence, 3rd)**, Text-Guided Colorization	*Inha Univ.*
2025.06	**Mayor's Award (Outstanding Idea, 3rd)**, Incheon Startup Competition	*Incheon*
2025.05	**Grand Prize (President's Award, 1st)**, MOONG IT Hackathon	*Inha Univ.*
2025.03	**Grand Prize (1st)**, Launch Pad Startup Competition	*Inha Univ.*
2024.11	**1st Place**, GDG Ideathon & Demo Day	*Inha Univ.*
2022.09	**Dean's Award (1st)**, Academic Excellence, AI Department	*Inha Univ.*
2021	**Academic Excellence Scholarship**, Academic achievement	*Inha Univ.*

Skills

Programming	Python, C++
Deep Learning	PyTorch, TensorFlow
RAG & Agents	LangChain, LangGraph, FAISS, Streamlit, MCP
3D & Design	Blender, Figma
Tools	Linux, Docker, Git, GitHub, SSH
Libraries	NumPy, Pandas, Matplotlib, Scikit-learn

Extracurricular Activity

AI Society: SINSA
Incheon, S. Korea

Vice President — *Jun. 2025 – Jun. 2026*

- Managed a university AI society under faculty supervision, overseeing programs in **Python, Data Analysis, Deep Learning, and Paper Study**.
- Organized internal **AI competitions** and structured **study curricula**, contributing to national awards in SW-centered University AI competitions.

RAG/Agent Study
Incheon, S. Korea

Study Leader — *Aug. 2025 – Present*

- Led a study group on **RAG and LLM-based Agents** using LangChain, LangGraph, and MCP.
- Combined theoretical exploration with hands-on **multimodal RAG/Agent implementation** and discussions of research and industrial applications.

Bishop–Deep Learning Study
Incheon, S. Korea

Study Member — *Mar. 2025 – Present*

- Participated in a year-long study of Bishop's *Deep Learning: Foundations and Concepts*, presenting **CNNs, diffusion models, and Transformers** through weekly peer teaching.

Start-up AI Agent Lab: UPFLOW
Incheon, S. Korea

Co-founding Member / AI Researcher — *Feb. 2025 – Oct. 2025*

- Led an **AI Agent startup** supported by the Pre-Startup Package and Lab-based Startup Program.
- Developed **enterprise AI Agent and MCP-based automation** for consulting and cosmetic OEM clients, gaining end-to-end experience from model development to commercialization.

CS231n Study (Stanford Course)
Incheon, S. Korea

Study Member — *Mar. 2025 – Jun. 2025*

- Participated in weekly discussions of **Stanford CS231n**, presenting CNN architectures and applications in computer vision.

Teaching

Environmental Data Analysis Mentoring
Incheon Metropolitan Office of Education

Mentor — *May. 2025 – Jun. 2025*

- Mentored middle and high school students in environmental data analysis using **KoNLPy, stopword removal, and Kkma** to analyze carbon-neutrality issues.
- Implemented a recycling-status **dashboard and object detection model** for recyclable-item classification.

Digital Green Workshop
Inha University

Mentor — *Oct. 2024*

- Mentored high school students in **AI-based environmental analysis**, guiding problem definition and **blockchain incentive** solutions with Inha University and the Incheon Metropolitan Office of Education.

EBS 1318 On-Coaching Program
Siheung, S. Korea

Mentor — *Jul. 2024 – Present*

- Provided individual mathematics and science tutoring in **equations, functions, and set theory**, emphasizing conceptual understanding.

Hwasung Mathematics Academy
Yongin, S. Korea

Teaching Assistant — *Mar. 2021 – Feb. 2022*

- Assisted curriculum design and personalized learning plans; taught **probability, statistics, and calculus** for college entrance preparation.
- Communicated with students and parents to coordinate academic goals and progress.

Certifications

ADsP
Korea Data Agency

Advanced Data Analytics Semi-Professional — *Mar. 22, 2024*